

BLOOD BANK & DONATION MANAGEMENT SYSTEM

REQUIREMENT ANALYSIS DOCUMENT

Project Information

Project Name: Blood Bank & Donation Management System

Project Type: Web Application

Testing Type: Manual Testing

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1. PROJECT OVERVIEW

The Blood Bank & Donation Management System is a web-based application designed to connect blood donors with patients who require blood. The system allows donors to register their information, patients to search for blood based on blood groups, and administrators to manage donor information and blood requests.

The objective of the application is to simplify blood donation management and help users quickly find suitable blood donors during emergencies.

2. MODULES IDENTIFIED

Module M001 – Home Page

Purpose:

Provide navigation to all major functionalities of the application.

Features:

- Navigation menu
- Latest updates section
- Blood donation awareness content
- Blood group information

- Donor awareness information

Expected Behaviour:

- Home page should load successfully.
- All navigation links should be accessible.
- Images and content should display correctly.

Module M002 – About Us

Purpose:

Provide information about the blood bank system.

Features:

- Organization information
- Blood bank description
- Awareness content

Expected Behaviour:

- Page should load successfully.
- Content should be readable.
- Images should display correctly.

Module M003 – Why Donate Blood

Purpose:

Educate users about blood donation benefits.

Features:

- Blood donation benefits
- Informational content
- Blood usage statistics

Expected Behaviour:

- Page should load correctly.

- Content should be visible.
- No broken images.

Module M004 – Become A Donor

Purpose:

Allow users to register as blood donors.

Input Fields:

- Full Name
- Mobile Number
- Email ID
- Age
- Gender
- Blood Group
- Address

Buttons:

- Submit

Expected Behaviour:

- Donor details should be accepted.
- Mandatory fields should be validated.
- Successful submission message should be displayed.

Business Rules:

- Full Name is mandatory.
- Mobile Number is mandatory.
- Age is mandatory.
- Gender is mandatory.
- Blood Group is mandatory.

- Address is mandatory.

Module M005 – Need Blood

Purpose:

Allow users to search for blood donors.

Input Fields:

- Blood Group
- Reason for Blood Requirement

Buttons:

- Search

Expected Behaviour:

- Users should be able to search by blood group.
- Matching donor information should be displayed.

Business Rules:

- Blood Group is mandatory.
- Search should return available donor records.

Module M006 – Contact Us

Purpose:

Allow users to communicate with the blood bank.

Input Fields:

- Full Name
- Phone Number
- Email Address
- Message

Buttons:

- Send Message

Expected Behaviour:

- User should be able to submit a message.
- Contact details should be visible.

Business Rules:

- Name is mandatory.
- Message should not be empty.

3. FUNCTIONAL REQUIREMENTS

FR001:

System shall display the home page successfully.

FR002:

System shall allow navigation between all pages.

FR003:

System shall allow donor registration.

FR004:

System shall validate mandatory fields.

FR005:

System shall allow blood search by blood group.

FR006:

System shall allow users to submit contact messages.

FR007:

System shall display blood donation awareness information.

4. NON-FUNCTIONAL REQUIREMENTS

NFR001:

Application should load within acceptable response time.

NFR002:

Application should be accessible through modern browsers.

NFR003:

Application UI should be responsive and user friendly.

NFR004:

System should maintain data consistency.

5. ASSUMPTIONS

- Internet connection is available.
- Database connection is active.
- Users access the application through supported browsers.
- Registered donor information is stored successfully.

6. RISKS

- Invalid user inputs.
- Duplicate donor registration.
- Missing mandatory validations.
- Database connectivity issues.

7. DELIVERABLES PLANNED

- Requirement Analysis Document
- Requirement Traceability Matrix (RTM)
- Test Scenarios
- Smoke Test Cases
- Functional Test Cases
- Integration Test Cases
- System Test Cases
- Regression Test Cases
- Defect Reports
- Jira Defect Tracking
- Test Summary Report